

Question ID: 3882ddf6

Question

To investigate the history of plate subduction—when one of Earth’s tectonic plates slides beneath another—Sarah M. Aaronson and colleagues compared ancient rocks from the Acasta Gneiss Complex in Canada to modern rocks. Using isotope analysis, the researchers found that Acasta rocks dating to about 4.02 billion years ago (bya) most strongly resemble modern rocks formed in a plume setting (an area in which hot rocks from Earth’s mantle flow upward into the crust). By contrast, they found that Acasta rocks dating to about 3.75 bya and 3.6 bya have an isotope composition that is similar to that of modern rocks formed in a subduction setting. Aaronson’s team therefore concluded that _____

Which choice most logically completes the text?

Answer

- A. subduction-like processes began occurring in some locations no later than 3.75 bya.
- B. subduction replaced mantle plume formation as the most common geological process by about 4.02 bya.
- C. the majority of the rocks in the Acasta Gneiss Complex formed through subduction.
- D. the rocks in the Acasta Gneiss Complex are of a more recent origin than scientists previously thought.